

ARI5118 - AI Usage Journal

CNN Feature Visualisation and DeepDream

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Section 1: Tools Declared

- **ChatGPT (GPT-5.5 Thinking):** Used for explaining concepts, improving wording, checking slide content, splitting code logic into separate files and generating UI ideas for the learning pack.
- **NotebookLM (Gemini 3 Flash for fast):** Used to explore uploaded study notes and academic sources, extract key points from references, help identify useful explanations for the topic, as well as possible questions for the quiz.
- **Claude AI (Sonnet 4.6 Adaptive):** Used to explore whether a CNN feature hierarchy element could be implemented in the simulator website and to generate initial reference code for the component. It was also used to suggest suitable colour palettes, while the final design and implementation were manually reviewed, edited and adapted.

Section 2: Notable Prompts

Prompt 1: Finding valid academic references

Prompt: “Hey ChatGPT I am currently working on another project and need references that are valid. The project itself is to create an education deck (slides, quiz) and a simulation. But for now references academic papers This is my topic CNN Feature Visualisation and DeepDream and the core focus Activation maximisation, gradient ascent on input, feature hierarchy, DeepDream amplification.”

Summary of response: The AI tool suggested relevant academic and technical sources, including papers and blogs on activation maximisation, feature visualisation, network interpretability, and DeepDream. This helped shape the reference list and gave direction for which papers were most relevant to the learning pack.

Prompt 2: Understanding and explaining activation maximisation

Prompt: “For activation maximisation, should the image be RGB or greyscale? What can I change or should I change because some filters are only returning a box of a single colour, or is that normal? (a test file of the implementation was attached)”

Summary of response: The response explained that activation maximisation can produce strange or abstract outputs, especially if the optimisation is weakly regularised or dominated by colour/texture patterns. It clarified that RGB is appropriate for models like VGG16 since it was trained on RGB ImageNet images. It also mentioned that single-colour outputs may indicate issues with objective choice, learning rate, regularisation or normalisation.

Prompt 3: Implementing simulator colour themes

Prompt: “I have this colour palette as of now for the simulator’s dark mode. Now, I wish to implement a way for users to change colour themes, how can I do this, using CSS and JS? The light theme should keep the same accents as the dark while the third theme should use the light as a base but switch the accents to different shades of green.”

Summary of response: The response suggested using CSS custom properties and a simple JavaScript theme switcher. Instead of rewriting styles for every component, the simulator could change a single attribute and update shared variables such as background, border, text and accent colours. This helped me implement theme switching in a cleaner and more maintainable way.

Prompt 4: Using NotebookLM to support quiz creation

Prompt: “So far this is what I have for my study notes in order to tackle my topic of CNN Feature Visualisation and DeepDream with a core Focus in Activation maximisation, gradient ascent on input, feature hierarchy, DeepDream amplification. (The quiz button built into NotebookLM was then used to generate reference question sets)”

Summary of response: NotebookLM used the uploaded study notes and sources to suggest possible quiz questions based on the main concepts in the learning pack. This was useful because I had already created some questions manually, but needed more ideas and different ways of testing the same material. The response helped me identify possible misconception-based questions and gave me a stronger starting point for building the final adversarial quiz.

Prompt 5: Generating presentation support materials in NotebookLM

Prompt / AI interaction: Using NotebookLM, I uploaded the required references and project sources, then used the built-in beta tools, specifically the slide deck and infographic generator to create presentation references for this project.

Summary of response: NotebookLM generated draft slide and infographic-style material from the uploaded sources. This was useful for getting visual ideas and possible ways to structure the presentation. However, some generated content was too broad or not reliable enough to use directly, so it was treated as reference material rather than final content.

Section 3: Five Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

The hardest part to explain was activation maximisation, especially the equations behind it. I initially understood the general idea that the input image is changed to maximise a neuron or filter activation, but explaining the mathematics behind it clearly was more difficult. For example, I had to understand what x , x_t , x_{t+1} , η , and $\frac{\partial a_i(x)}{\partial x}$ represented in the update rule. This was challenging since past projects usually dive into practical example how to code it and move on.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

Sometimes, the AI tools gave answers that looked useful, but were either too detailed or slightly out of scope. This happened most often when using NotebookLM. I also noticed that AI-generated slide content tends to contain more obvious hallucinations. To fix this, I checked the AI outputs against my study notes, and when something seemed unclear or possibly incorrect, I

used other tools such as ChatGPT and Claude to double-check it. I then simplified the wording and removed details that were unnecessary or outside the scope of this project.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

AI wasted my time when I was working on the simulator's JavaScript and CSS. Some suggestions added new functions, extra variables and unnecessary style changes instead of improving and modularising the code that was already present. This ended up being a waste of time since code later had to be removed or simplified.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

AI genuinely made the work faster when I was designing the simulator interface and presentation style. I am not very confident when designing UI systems from scratch, so AI helped by suggesting colour palettes, CSS ideas and visual references for how the simulator and slides could look. This gave me a clearer starting point instead of designing everything from scratch. The final design was still reviewed and adapted manually, but AI made the early design process much quicker and helped the final learning pack look more polished.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

One question I still do not feel fully confident answering is how much trust should be placed in feature visualisation outputs as well as its use case. These techniques are useful for exploring what a CNN may be responding to, but the results are still affected by choices such as the target layer and the settings used and how the viewer interprets the image. Even after building the learning pack, Even after building the learning pack, I am still unsure how useful these techniques are outside of teaching or artistic inspiration, especially when more stable techniques may be better suited for real-world decision-making tasks.

A Prompt Images

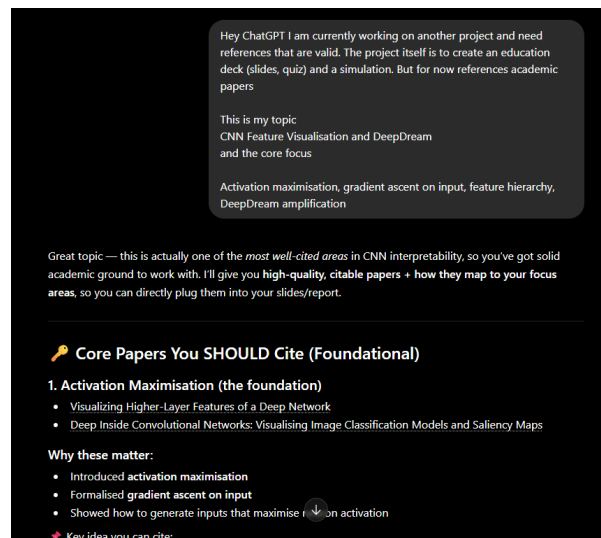


Figure 1: Prompt 1: Finding valid academic references

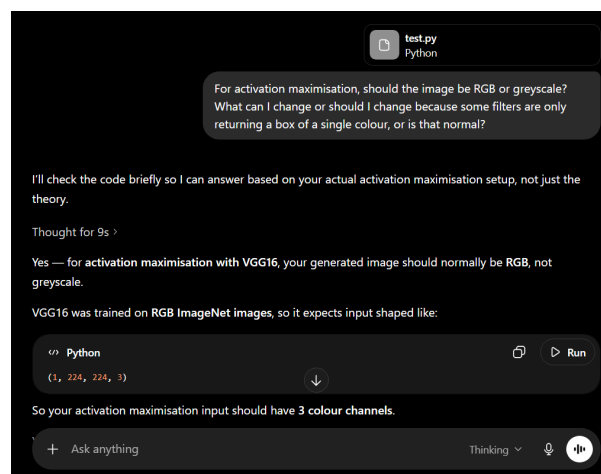


Figure 2: Prompt 2: Understanding and explaining activation maximisation

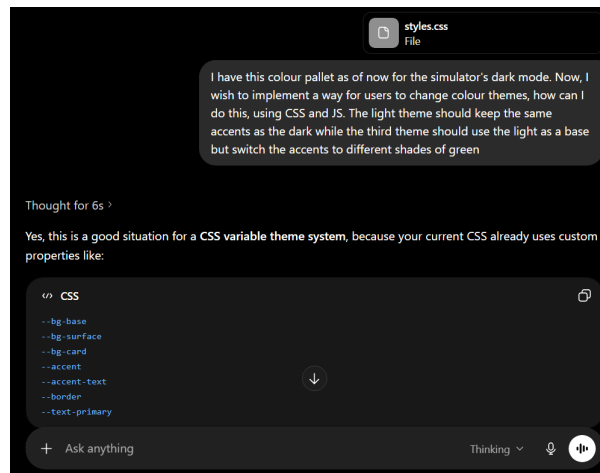


Figure 3: Prompt 3: Implementing simulator colour themes

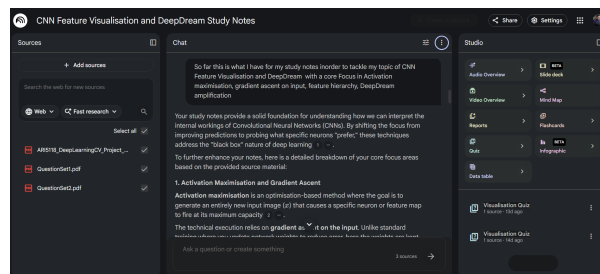


Figure 4: Prompt 4: Using NotebookLM to support quiz creation

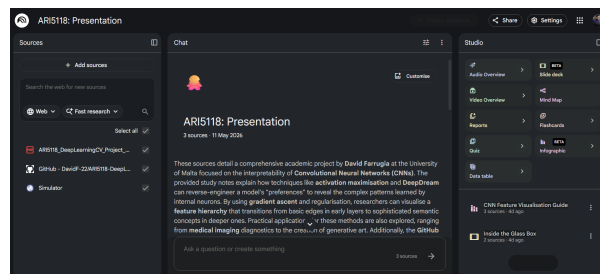


Figure 5: Prompt 5: Implementing simulator colour themes